

# PE-EC801B — Fiber Optic Communication

Semester VIII • 3L:0T:0P • 3 Credits

## Module 3 Exam Notes

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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## 1. Optical Sources — LED

### Light-Emitting Diode (LED)

An LED is a forward-biased p-n junction semiconductor device that emits light by **spontaneous emission** when electrons recombine with holes. LEDs for fiber optics operate in the 850 nm, 1310 nm, or 1550 nm windows.

### 1.1 Surface-Emitting vs Edge-Emitting LED

- **Surface-Emitting LED (SLED / Burrus LED):** Emits light perpendicular to the junction plane. Large emitting area ( $\sim 50 \mu\text{m}$  diameter), Lambertian emission pattern, easy to couple to MMSI/MMGI fibers. Modulation bandwidth  $\sim 20\text{--}100$  MHz.
- **Edge-Emitting LED (ELED):** Emits light parallel to the junction from the cleaved edge of the active stripe. Narrower emission cone, better coupling to single-mode or small-core multimode fibers. Modulation bandwidth  $\sim 100\text{--}400$  MHz. Output spectrum FWHM  $\sim 30\text{--}80$  nm.

### LED Output and Modulation Bandwidth

$$P_{opt} \propto I_d \cdot \eta_{ext}, \quad P(\omega) = \frac{P(0)}{\sqrt{1 + (\omega\tau_r)^2}}$$

Modulation bandwidth:  $f_{3dB} = \frac{\sqrt{3}}{2\pi\tau_r}$ , where  $\tau_r$  is the minority carrier radiative lifetime ( $\sim 2\text{--}10$  ns).

**Advantages of LEDs:** Simple construction, lower cost, longer lifetime ( $> 10^7$  hours), linear L-I characteristic, no threshold, operable without temperature control, immune to catastrophic failure.

**Disadvantages:** Broad output spectrum (large material dispersion), lower output power, slower modulation compared to laser diodes.

## 2. Optical Sources — Laser Diodes

### Laser Diode

A **laser diode** (LD) is a semiconductor p-n junction in which **stimulated emission** dominates above a threshold current  $I_{th}$ . It produces coherent, narrow-spectrum, high-power output suitable for SMF and high-speed links.

### 2.1 Fabry-Perot (FP) Laser

Uses two cleaved facets as reflective mirrors forming a Fabry-Perot cavity. Oscillation occurs at wavelengths satisfying the resonance condition  $L = m\lambda/(2n)$  ( $m$  integer). FP lasers are **multi-longitudinal-mode**; spectrum is a comb of modes separated by  $\Delta\lambda = \lambda^2/(2nL)$ .

## 2.2 DFB (Distributed Feedback) Laser

A Bragg grating is integrated into the active layer. Feedback is distributed along the cavity. Only the mode satisfying the Bragg condition  $\lambda_B = 2n\Lambda$  (where  $\Lambda$  is the grating period) is amplified. DFB lasers are **single-longitudinal-mode**, essential for WDM systems.

## 2.3 Rate Equations and Threshold Current

### Laser Rate Equations and L-I Characteristic

Carrier density  $N$  and photon density  $S$ :

$$\frac{dN}{dt} = \frac{\eta_i I}{eV} - \frac{N}{\tau_n} - Gg_0(N - N_0)S$$

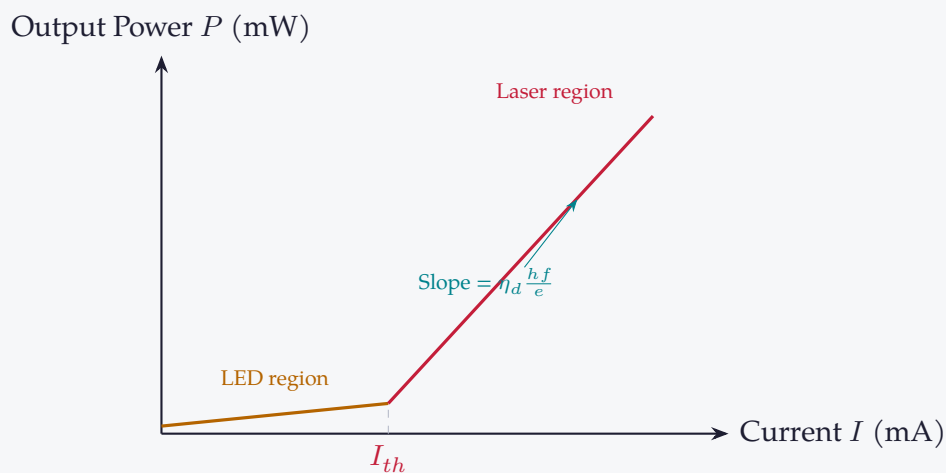
$$\frac{dS}{dt} = Gg_0(N - N_0)S + \frac{\beta N}{\tau_n} - \frac{S}{\tau_p}$$

Threshold current:  $I_{th}$  is where gain = loss (photon density starts growing exponentially).

$$P_{out} = \eta_d \cdot \frac{hf}{e}(I - I_{th}), \quad I > I_{th}$$

where  $\eta_d$  is differential quantum efficiency,  $hf$  is photon energy.

### L-I Curve of a Laser Diode



## 3. Photodetectors — PIN Diode and APD

### Photodetector — Responsivity and Quantum Efficiency

A **photodetector** converts optical power into electrical current via the photoelectric effect. **Responsivity**  $\mathcal{R}$  (A/W) is the photocurrent per unit optical power. **Quantum efficiency**  $\eta$  is the fraction of incident photons that generate electron-hole pairs.

**Responsivity Formula**

$$\mathcal{R} = \frac{I_p}{P_{opt}} = \frac{\eta e \lambda}{hc} \quad (\text{A/W})$$

where  $\eta$  = quantum efficiency,  $e$  = electron charge ( $1.6 \times 10^{-19}$  C),  $\lambda$  = wavelength (m),  $h$  = Planck's constant ( $6.626 \times 10^{-34}$  J·s),  $c$  = speed of light.

Also:  $\mathcal{R} = \eta e / (hf)$  where  $f = c/\lambda$  is optical frequency.

**3.1 PIN Photodiode**

A **PIN diode** has a lightly doped intrinsic (I) layer between heavily doped P and N regions. The wide depletion region extends across the I layer under reverse bias, allowing efficient absorption of photons over a large volume.

- **Structure:**  $P^+ / I(\text{intrinsic}) / N^+$ , reverse-biased.
- **Bandwidth:** Limited by transit time  $\tau_{tr} = d/v_d$  and RC time constant.  $f_{3dB} \approx 1/(2\pi\tau_{tr})$ . Bandwidths up to tens of GHz are achievable.
- **Responsivity:** Typically  $\mathcal{R} \approx 0.6\text{--}0.9$  A/W at 1300–1550 nm (InGaAs PIN).

**3.2 Avalanche Photodiode (APD)**

An APD operates under high reverse bias where photogenerated carriers gain sufficient energy to ionize more carriers through **impact ionization** — the avalanche effect. This provides internal current gain  $M$ .

**APD Gain, Noise, and SNR**

APD multiplication gain:  $M = I_{APD} / (I_{primary})$ .

Excess noise factor:  $F(M) \approx k_A M + (1 - k_A)(2 - 1/M)$  (McIntyre approximation), where  $k_A$  is the ionization ratio ( $k_A = \alpha_h / \alpha_e$ ).

For InGaAs APD:  $M$  typically 10–30;  $k_A \approx 0.4\text{--}0.7$ .

**3.3 Noise in Photodetectors**

- **Shot noise:** Due to the quantum nature of photon detection.  $\langle i_s^2 \rangle = 2eI_p B$  (PIN);  $\langle i_s^2 \rangle = 2eI_p M^2 F(M) B$  (APD), where  $B$  is bandwidth.
- **Thermal noise (Johnson noise):** Due to random thermal agitation of electrons in the load resistor  $R_L$ .  $\langle i_T^2 \rangle = 4k_B T B / R_L$ , where  $k_B = 1.38 \times 10^{-23}$  J/K.
- **SNR:**

$$\text{SNR}_{PIN} = \frac{(\mathcal{R}P_{opt})^2}{2e\mathcal{R}P_{opt}B + 4k_B T B / R_L}$$

$$\text{SNR}_{APD} = \frac{(M\mathcal{R}P_{opt})^2}{2eM^2 F(M)\mathcal{R}P_{opt}B + 4k_B T B / R_L}$$

**3.4 Optical Receivers — Pre-Amplifier Types**

- **High-impedance (HZ) pre-amplifier:** Large  $R_L$  minimizes thermal noise, but limits bandwidth due to RC roll-off. Equalizer needed. Best sensitivity.

- **Transimpedance (TIA) pre-amplifier:** Uses feedback resistor  $R_f$  to convert photocurrent to voltage. Flat frequency response up to higher bandwidth with good sensitivity. Preferred in practice.

## 4. Optical Link Design — BER, Q-Factor, and Receiver Sensitivity

**Optical link design** is the process of ensuring that a fiber optic system meets both the **power budget** (sufficient received power for the target BER after accounting for all losses) and the **dispersion budget** (pulse broadening stays within the allowed bit-period fraction). The power budget gives:  $P_{tx} - (\text{total losses}) \geq P_{rx,min}$ , where losses include fiber attenuation ( $\alpha L$ ), splice losses, connector losses, and a system margin ( $\sim 3\text{--}6\text{ dB}$ ). The dispersion budget requires:  $\sigma_{total} \leq T_b/(4\sqrt{2})$  (Gaussian pulse limit), where  $\sigma_{total} = \sqrt{\sigma_{modal}^2 + \sigma_{chromatic}^2 + \sigma_{source}^2}$ .

### BER and Q-Factor

**Bit Error Rate (BER)** is the probability that a received bit is in error. The **Q-factor** is the signal-to-noise amplitude ratio that directly maps to BER. For a Gaussian noise model:  $\text{BER} = \frac{1}{2}\text{erfc}(Q/\sqrt{2})$ .

### Link Design Equations

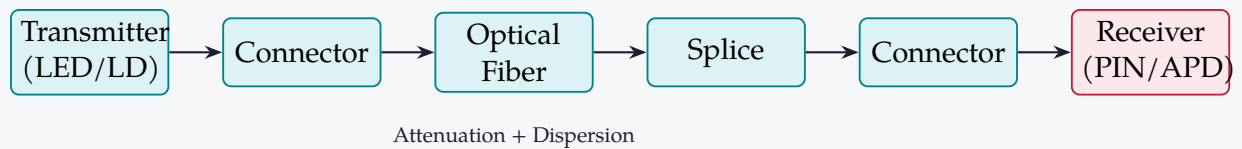
|                                            |                                                                                                                      |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>Q-factor</b>                            | $Q = \frac{I_1 - I_0}{\sigma_1 + \sigma_0}$ ( $I_1, I_0$ : mean currents for 1, 0)                                   |
| <b>BER vs Q</b>                            | $\text{BER} \approx \frac{1}{2}\text{erfc}\left(\frac{Q}{\sqrt{2}}\right) \approx \frac{\exp(-Q^2/2)}{Q\sqrt{2\pi}}$ |
| <b>BER = <math>10^{-9}</math> requires</b> | $Q \approx 6$                                                                                                        |
| <b>Quantum limit</b>                       | $\text{BER}_{QL} = \frac{1}{2}e^{-\eta N_p}$ , minimum $N_p \approx 10$ photons/bit                                  |
| <b>Receiver sensitivity</b>                | Minimum $P_{min}$ (dBm) for target BER                                                                               |

### 4.1 Power Penalties

Power penalties are additional receiver power margins required due to system imperfections beyond the AWGN model.

| Penalty Source              | Cause                                                       | Typical Range                  |
|-----------------------------|-------------------------------------------------------------|--------------------------------|
| <b>Modal noise</b>          | Mode partition noise in MMF; speckle pattern changes        | 0.5–1 dB                       |
| <b>Extinction ratio</b>     | Non-zero optical power for “0” bit ( $r = P_0/P_1 \neq 0$ ) | 1–2 dB (for $r = 0.1$ )        |
| <b>Timing jitter</b>        | Random deviation in bit timing $\Rightarrow$ ISI            | 0.5–2 dB                       |
| <b>Chromatic dispersion</b> | Pulse broadening causes ISI at high bit rate                | Depends on $D \cdot B \cdot L$ |
| <b>Reflection noise</b>     | Multiple reflections create interferometric noise           | 0.5–1 dB                       |

#### Block Diagram — Complete Fiber Optic Link



## Quick Revision — Module 3

Optical Sources, Detectors, and Link Design | PE-EC801B

### Key Formulas at a Glance

|                    |                                                        |
|--------------------|--------------------------------------------------------|
| Responsivity       | $\mathcal{R} = \eta e \lambda / (hc)$ (A/W)            |
| Shot noise (PIN)   | $\langle i_s^2 \rangle = 2e\mathcal{R}P_{opt}B$        |
| Thermal noise      | $\langle i_T^2 \rangle = 4k_B T B / R_L$               |
| LED modulation BW  | $f_{3dB} = \sqrt{3} / (2\pi\tau_r)$                    |
| Laser output power | $P_{out} = \eta_d (hf/e)(I - I_{th})$ ( $I > I_{th}$ ) |
| Q-factor           | $Q = (I_1 - I_0) / (\sigma_1 + \sigma_0)$              |
| BER (Gaussian)     | $BER = \frac{1}{2} \text{erfc}(Q/\sqrt{2})$            |
| BER = $10^{-9}$    | $Q \approx 6$                                          |
| APD SNR            | $\propto M^2 / F(M)$ (optimum $M$ exists)              |

### One-Line Definitions

- **Spontaneous emission:** Random photon emission by an excited electron without external stimulus (LED).
- **Stimulated emission:** A passing photon triggers an identical photon from an excited state (Laser).
- **Threshold current  $I_{th}$ :** The bias current above which laser gain exceeds cavity loss.
- **DFB laser:** Distributed Feedback laser; single-mode due to integrated Bragg grating.
- **Responsivity  $\mathcal{R}$ :** Photocurrent generated per watt of incident optical power (A/W).
- **Quantum efficiency  $\eta$ :** Fraction of photons that create an electron-hole pair.
- **APD gain  $M$ :** Internal multiplication of photocurrent by impact ionization.
- **Q-factor:** SNR amplitude ratio;  $Q = 6$  gives  $BER = 10^{-9}$ .
- **Transimpedance amp:** Pre-amplifier with feedback  $R_f$  giving flat bandwidth and good sensitivity.

### Frequently Asked / Viva Questions

#### Q1. What is the difference between an LED and a Laser diode?

LED uses spontaneous emission; broad spectrum; no threshold; lower power. Laser diode uses stimulated emission above  $I_{th}$ ; narrow coherent spectrum; higher power; higher modulation speed.

## Frequently Asked / Viva Questions (contd.)

**Q2. What is the threshold current in a laser diode?**

It is the minimum bias current at which the gain of the active medium equals the total cavity losses, causing the onset of lasing (stimulated emission dominates).

**Q3. Why is a DFB laser preferred for WDM systems?**

A DFB laser emits a single longitudinal mode (due to the Bragg grating), giving a very narrow linewidth ( $<1$  MHz). This is essential to prevent channel overlap in dense WDM systems.

**Q4. Define responsivity and derive the formula  $\mathcal{R} = \eta e \lambda / hc$ .**

$\mathcal{R} = I_p / P_{opt}$ . Each photon of energy  $hf = hc/\lambda$  generates  $\eta$  electron-hole pairs, contributing charge  $\eta e$ . Per second,  $P_{opt}/(hf)$  photons arrive, giving  $I_p = \eta e P_{opt}/(hf) = \eta e \lambda P_{opt}/(hc)$ , so  $\mathcal{R} = \eta e \lambda / (hc)$ .

**Q5. What is the advantage of APD over PIN diode?**

APD provides internal current gain  $M$  through avalanche multiplication, improving receiver sensitivity (lower minimum detectable power). However, it introduces excess noise  $F(M)$  and requires higher bias voltage.

**Q6. What is shot noise? When is it dominant?**

Shot noise arises from the discrete quantum nature of photon detection.  $\langle i_s^2 \rangle = 2eI_p B$ . It dominates over thermal noise at high optical power or when the load resistor is large.

**Q7. Define Q-factor and state its relation to BER.**

$Q = (I_1 - I_0)/(\sigma_1 + \sigma_0)$ . For Gaussian noise:  $\text{BER} = \frac{1}{2} \text{erfc}(Q/\sqrt{2})$ .  $\text{BER} = 10^{-9}$  requires  $Q \approx 6$ .

**Q8. What is the quantum limit of detection?**

The absolute minimum BER is set by the Poisson statistics of photon arrival:  $\text{BER} = \frac{1}{2} e^{-\eta N_p}$ . The quantum limit is  $\sim 10$  photons/bit for ideal detection.

**Q9. What is extinction ratio penalty?**

When the laser emits non-zero power for a "0" bit (extinction ratio  $r = P_0/P_1 \neq 0$ ), the eye opening reduces, requiring extra received power to maintain the same BER.  $\text{Penalty} = 10 \log_{10} \left( \frac{1+r}{1-r} \right)$ .

**Q10. Compare surface-emitting and edge-emitting LEDs.**

SLED: wide emitting area, Lambertian pattern, easy coupling to MMF, BW  $\sim 100$  MHz. ELED: narrow emission cone, better coupling to SMF or small MMF, higher BW  $\sim 400$  MHz, higher power.

**Q11. What is the difference between high-impedance and transimpedance pre-amplifiers?**

HZ: large  $R_L$  minimizes thermal noise; limited bandwidth needing equalization. TIA: uses feedback  $R_f$ ; flat bandwidth; moderate noise; preferred for practical high-speed receivers.

## LED vs Laser Diode — Comparison

| Parameter               | LED             | Laser Diode         |
|-------------------------|-----------------|---------------------|
| Emission type           | Spontaneous     | Stimulated          |
| Coherence               | Incoherent      | Coherent            |
| Spectral width          | 30–80 nm        | <1 nm (DFB)         |
| Threshold               | None            | $I_{th}$ required   |
| Modulation BW           | ~100–400 MHz    | >10 GHz             |
| Output power            | Low (mW)        | High (>10 mW)       |
| Temperature sensitivity | Low             | High                |
| Cost                    | Low             | High                |
| Application             | Short-haul, MMF | Long-haul, SMF, WDM |